



2026 Released Tests

Aligned to the Standards

CONTENT BUILDER FOR THE PLC

Science Biology

The diagram illustrates the relationship between Student Expectation, Data Analysis, and Descriptive Measures. It features a central table with three main sections: Student Expectation, Data Analysis, and Descriptive Measures. Arrows indicate the flow of information and the relationship between these sections.

Student Expectation | SE Type
 DISCUSS: How many questions for this Student Expectation were asked over the past years? Which parts were assessed?

[Cluster/Genre]: [Subcluster]
[Student Expectation] [type]

Data Analysis
 DISCUSS and NOTE: Review the state-level answer choice data, item type, and points possible for each item.
 COMPLETE: Add local data for the item.

Descriptive Measures
 DISCUSS and NOTE:
 • **Item Complexity:** What level of thinking is required to answer the question correctly?
 • **Stimulus:** How many different stimuli (visuals) were used to assess this student expectation?
 • **Academic Vocabulary:** How will we help students learn and use the important words in context?
 • **Common Error Patterns:** What are the most common error patterns?
 INSTRUCTIONAL IMPLICATIONS: How will we adapt instruction to help students improve understanding and application of this student expectation?

Item Complexity

- Basic application –requires foundational knowledge and application of content
- Think critically –requires critical thinking about the content (M, Sc, SS) or overall meaning (RLA)
- Think critically in complex contexts –requires deep analysis or critical thinking in complex contexts

* Correct Answer

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- **Content Area:** Science
- **Grade:** Biology
- **Clusters:** Cell Structure and Biochemistry, Organism Growth and Cell Differentiation, Mechanisms of Genetics, Heredity and Diversity, Evolutionary Theory, Evolutionary Mechanisms, Energy Conversions in Organisms, Interactions Among Biological Systems, Interdependence and Ecosystem Stability
- **Student Expectations:** B.5(A), B.5(B), B.5(D), B.6(A), B.6(C), B.6(B), B.7(A), B.7(B), B.7(C), B.8(B), B.8(A), B.9(A), B.9(B), B.10(C), B.10(A), B.10(B), B.10(D), B.11(A), B.11(B), B.12(A), B.12(B), B.13(A), B.13(B), B.13(D), B.13(C)
- **Years:** 2026

Cell Structure and Biochemistry

B.5 Biological structures, functions, and processes. The student knows that biological structures at multiple levels of organization perform specific functions and processes that affect life.

Cell Structure and Biochemistry: Cellular Complexity

B.5(B) compare and contrast prokaryotic and eukaryotic cells, including their complexity, and compare and contrast scientific explanations for cellular complexity (Readiness)

2026 – Q8

A student views four cells under a light microscope and records some observations in the table shown.

Which observation in the table can be used to identify one of the cells as a prokaryote?

Select **ONE** correct answer.

Specimen	Flagellum Present?	Nucleus Present?	Chloroplast Present?	Cell Wall Present?
Cell 1	Yes	Yes	No	No
Cell 2	No	Yes	No	No
Cell 3	Yes	No	No	Yes
Cell 4	No	Yes	Yes	Yes

*Correct Answer ("No" under "Nucleus Present?")

Data Analysis

Item	State	Local
Full Credit	66	
No Credit	34	
Item Type	Hot Spot	
Points	1	

Descriptive Measures

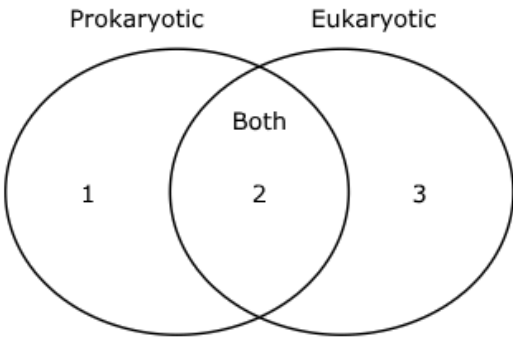
- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

Cell Structure and Biochemistry: Cellular Complexity

B.5(B) compare and contrast prokaryotic and eukaryotic cells, including their complexity, and compare and contrast scientific explanations for cellular complexity (Readiness)

2026 – Q26

The Venn diagram compares prokaryotic and eukaryotic cells.



Which table correctly completes the Venn diagram?

A

1	Linear chromosome
2	Nucleic acids
3	Plasmid

B

1	Linear chromosome
2	Plasmid
3	Nucleic acids

C

1	Plasmid
2	Linear chromosome
3	Nucleic acids

D

1	Plasmid
2	Nucleic acids
3	Linear chromosome

*Correct Answer (D)
^highly chosen incorrect answer (C)

Data Analysis

Item	State	Local
A		
B		
C	^	
D*	33	
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

Organism Growth and Cell Differentiation

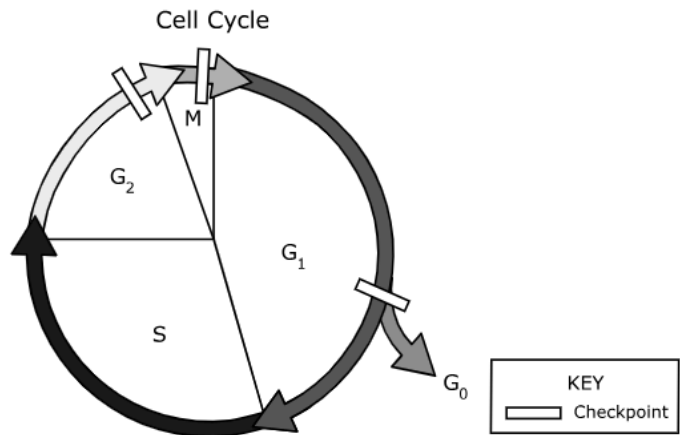
B.6 Biological structures, functions, and processes. The student knows how an organism grows and the importance of cell differentiation.

Organism Growth and Cell Differentiation: Cell Cycle

B.6(A) explain the importance of the cell cycle to the growth of organisms, including an overview of the stages of the cell cycle and deoxyribonucleic acid (DNA) replication models (Supporting)

2026 – Q18

A diagram of the cell cycle is shown.



When the checkpoints are functioning properly, how do cells respond to them?

- ☐ (A) An abnormal cell proceeds, and a normal cell undergoes apoptosis.
- ☐ (B) An abnormal cell undergoes apoptosis, and a normal cell proceeds.
- ☐ (C) An abnormal cell and a normal cell both undergo apoptosis.
- ☐ (D) An abnormal cell and a normal cell both proceed.

* Correct Answer (B)
^highly chosen incorrect answer (A)

Data Analysis		
Item	State	Local
A	^	
B*	71	
C		
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

IQ Analysis Investigating the Question	B.6(B)	RC 1
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Organism Growth and Cell Differentiation: Cell Differentiation			
B.6(B) explain the process of cell specialization through cell differentiation, including the role of environmental factors (Supporting)			
2026 – Q15			
Somatic cells in humans usually contain 23 pairs of chromosomes. Which sentence BEST describes why human cells can have different functions?			
A Different genes are expressed in various types of cells. B Gene regulation malfunctions in some cells, causing differentiation to occur. C Nucleic acids are the same in all cells, so gene expression occurs only due to mutations. D RNA controls gene expression as proteins are synthesized.			

IQ Analysis | Investigating the Question

B.6(C)

RC 1

Organism Growth and Cell Differentiation: Cell Cycle

B.6(C) relate disruptions of the cell cycle to how they lead to the development of diseases such as cancer (Readiness)

2026 – Q11

Which statement describes how cells **MOST LIKELY** become cancerous?

- ☐ A ATP is no longer needed for cell division since division is now self-regulated.
- ☐ B Unregulated cell division occurs because of mutations in the DNA.
- ☐ C DNA replication is omitted because of changes in the genetic code.
- ☐ D Damage to DNA molecules results in improved responses to cell cycle checkpoints.

*Correct Answer (B)

^highly chosen incorrect answer (D)

Data Analysis

Item	State	Local
A		
B*	71	
C		
D	^	
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

Organism Growth and Cell Differentiation: Cell Cycle

B.6(C) relate disruptions of the cell cycle to how they lead to the development of diseases such as cancer (Readiness)

2026 – Q22

The model shows some of the events that take place when a cell reaches the G₂ checkpoint in the cell cycle. How should the model be completed to show the **MOST LIKELY** outcome for this cell?

Move the correct answer to the box. Not all answers will be used.

- Apoptosis takes place.
- Cell returns to G₁ checkpoint.
- Cell turns into a tumor.
- New DNA is synthesized.

DNA damage is found. → DNA repair is attempted. → DNA repair fails. →

*Correct Answer (Apoptosis takes place)

Data Analysis

Item	State	Local
Full Credit	42	
No Credit	58	
Item Type	Drag and Drop	
Points	1	

Descriptive Measures

- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

Mechanisms of Genetics

B.7 Mechanisms of genetics. The student knows the role of nucleic acids in gene expression.

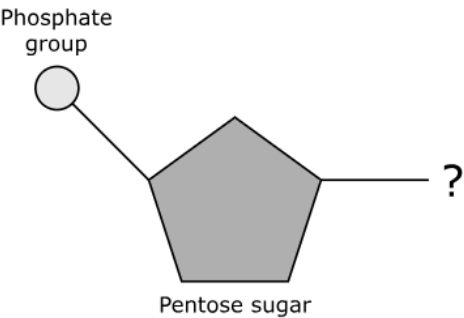
IQ Analysis Investigating the Question	B.7(A)	RC 2
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Mechanisms of Genetics: DNA

B.7(A) identify components of DNA, explain how the nucleotide sequence specifies some traits of an organism, and examine scientific explanations for the origin of DNA (Supporting)

2026 – Q5

A DNA nucleotide has three components. This model of a nucleotide is missing one of these components.



Identify the missing nucleotide component, **AND** explain the significance of this component in determining the traits of an organism.
Read the information carefully. Then enter your response in the box provided.

B *I* U *T_x* | | |

Chars 0/475

*Correct Answer (See Scoring Guide)

Data Analysis		
Item	State	Local
Full Credit	6	
No Credit	68	
Partial Credit	26	
Item Type	Short Constructed Response	
Points	2	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

IQ Analysis Investigating the Question	B.7(B)	RC 2
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Mechanisms of Genetics: Gene Expression

B.7(B) describe the significance of gene expression and explain the process of protein synthesis using models of DNA and ribonucleic acid (RNA) (Supporting)

2026 – Q32

Anthocyanin is a pigment that gives some plants red or blue coloration. Scientists examined the production of anthocyanin in fresh apples stored under different environmental conditions. The graph shows the amount of anthocyanin pigment per gram of fresh apples.

Anthocyanin Production under Different Environmental Conditions

Temperature (°C)	Without UV radiation (micrograms/gram)	With UV radiation (micrograms/gram)
17	4.5	6.5
27	4.0	10.5

Source: Ban, et al., *Plant and Cell Physiology*, 2007.

Based on the graph, which statement correctly describes the conditions **MOST LIKELY** to produce the proteins involved in the gene expression of anthocyanin?

A

Messenger RNA is most active when UV radiation is absent and when temperatures are warm.

B

Messenger RNA is most active when UV radiation is present and when temperatures are cool.

C

Messenger RNA is most active when UV radiation is present and when temperatures are warm.

D

Messenger RNA is most active when UV radiation is absent and when temperatures are cool.

* Correct Answer (C)

^highly chosen incorrect answer (B)

Data Analysis

Item	State	Local
A		
B	^	
C*	69	
D		
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

Item Complexity

Basic application

Think critically

Think critically in complex contexts

Stimulus

Academic Vocabulary

Common Error Patterns

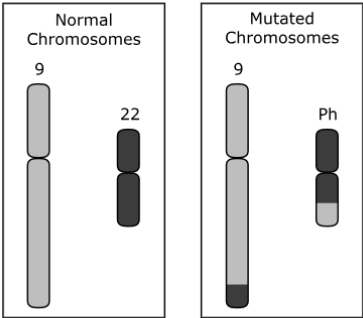
IQ Analysis Investigating the Question	B.7(C)	RC 2
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Mechanisms of Genetics: Mutations

B.7(C) identify and illustrate changes in DNA and evaluate the significance of these changes (Readiness)

2026 – Q9

The Philadelphia chromosome (Ph) is a chromosomal abnormality that results in cancer in humans. This occurs when portions of chromosome 9 and chromosome 22 are switched, as shown in the image.



Which type of chromosomal mutation must occur in order to form the Philadelphia chromosome?

- ☒ A Translocation
- ☐ B Deletion
- ☐ C Insertion
- ☐ D Nondisjunction

***Correct Answer (A)**
^highly chosen incorrect answer (C)

Data Analysis		
Item	State	Local
A*	69	
B		
C	^	
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

Mechanisms of Genetics: Mutations

B.7(C) identify and illustrate changes in DNA and evaluate the significance of these changes (Readiness)

2026 – Q25

A student is creating a table with an example of a DNA sequence that experienced a mutation and a description of this mutation. Which table provides the **BEST** example and description of a frameshift mutation?

A

Original DNA	3' CGT TAT GCT T 5'
Mutated DNA	3' CTC GTA TTG T 5'
Mutation Description	A section of DNA breaks away and reattaches to the chromosome in the reverse order without affecting the resulting protein.

B

Original DNA	3' CGT TAT GCT T 5'
Mutated DNA	3' CGT GTA TGC TT 5'
Mutation Description	A single base is inserted, which could result in a protein with a different amino acid sequence and shape.

C

Original DNA	3' CGT TAT GCT T 5'
Mutated DNA	3' CGT AAT GCT T 5'
Mutation Description	One base is substituted for another, which could result in a stop codon and a protein with a different amino acid sequence and shape.

D

Original DNA	3' CGT TAT GCT T 5'
Mutated DNA	3' CGT TAA GCT T 5'
Mutation Description	A single base is changed and codes for the same amino acid without affecting the resulting protein.

*Correct Answer (B)
^highly chosen incorrect answer (C)

Data Analysis		
Item	State	Local
A		
B*	39	
C	^	
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Heredity and Diversity

B.8 Mechanisms of genetics. The student knows the role of nucleic acids and the principles of inheritance and variation of traits in Mendelian and non-Mendelian genetics.

B.8(A) analyze the significance of chromosome reduction, independent assortment, and crossing-over during meiosis in increasing diversity in populations of organisms that reproduce sexually (Supporting)

The processes of crossing-over and independent assortment occur during in . These processes genetic diversity by producing gametes that allow some of the individuals in the population to survive following an environmental change such as a fire.

***Correct Answer (meiosis, fungi but not in bacteria, increase, unique)**

- **Item Complexity**
 - Basic application
 - Think critically
 - Think critically in complex contexts
- **Stimulus**
- **Academic Vocabulary**
- **Common Error Patterns**

IQ Analysis Investigating the Question	B.8(B)	RC 2
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Heredity and Diversity: Genetic Diversity		
B.8(B) predict possible outcomes of various genetic combinations using monohybrid and dihybrid crosses, including non-Mendelian traits of incomplete dominance, codominance, sex-linked traits, and multiple alleles (Readiness)		
2026 – Q4 Roan cattle have red-and-white fur evenly distributed on the hide and have the genotype <i>RW</i>. A roan cow is crossed with a red bull having the genotype <i>RR</i>. What is the probability of the genotype <i>RW</i> appearing in the offspring? A 0% B 25% C 50% D 100% *Correct Answer (C) ^highly chosen incorrect answer (B)	Data Analysis	
	Item	StateLocal
	A	
	B	^
	C*	79
	D	
	Item Type	Multiple Choice
	Points	1
	Descriptive Measures	
	- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns	

Heredity and Diversity: Genetic Diversity

B.8(B) predict possible outcomes of various genetic combinations using monohybrid and dihybrid crosses, including non-Mendelian traits of incomplete dominance, codominance, sex-linked traits, and multiple alleles (Readiness)

2026 – Q20

A gardener is growing sunflowers, which have an incomplete dominance inheritance pattern. The sunflowers can produce red, orange, and yellow flowers. The gardener crosses a red sunflower with the genotype *RR* with a yellow sunflower with the genotype *rr*.

Which outcome is the **MOST LIKELY** result of this cross?

- Ⓐ One-quarter of the sunflowers will be yellow.
- Ⓑ Half the sunflowers will be red.
- Ⓒ Three-fourths of the sunflowers will have both red and yellow petals.
- Ⓓ All the sunflowers will be orange.

*Correct Answer (D)
^highly chosen incorrect answer (C)

Data Analysis		
Item	State	Local
A		
B		
C	^	
D*	57	
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

Evolutionary Theory

B.9 Biological evolution. The student knows evolutionary theory is a scientific explanation for the unity and diversity of life that has multiple lines of evidence.

Evolutionary Theory: Evolution

B.9(A) analyze and evaluate how evidence of common ancestry among groups is provided by the fossil record, biogeography, and homologies, including anatomical, molecular, and developmental (Supporting)

2026 – Q1

The table shows partial amino acid sequences for a molecule produced by several species.

	Amino Acid					
Species	1	2	3	4	5	6
Unknown	Tyr	Val	Pro	Leu	Phe	Ser
R	Trp	Leu	Val	Trp	His	Ile
S	Ile	Asn	Arg	Leu	Asn	Ser
T	Tyr	Val	His	Leu	Asn	Ile
U	Ile	Leu	Arg	Cys	Thr	Ser

Based on the amino acid sequences shown, which species is **MOST** closely related to the unknown species?

- A

 Species R
- B

 Species S
- C

 Species T
- D

 Species U

*Correct Answer (C)
^highly chosen incorrect answer (B)

Data Analysis

Item	State	Local
A		
B	^	
C*	87	
D		
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

IQ Analysis Investigating the Question	B.9(B)	RC 3
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Evolutionary Theory: Evolution		
B.9(B) examine scientific explanations for varying rates of change such as gradualism, abrupt appearance, and stasis in the fossil record (Readiness)		
2026 – Q6 Bryozoans are a group of coral-like marine organisms. The well-preserved fossil record of bryozoans shows that one species first appeared around 140 million years ago. This species remained unchanged for close to 40 million years, and then there was a major diversification. This was followed by another long period of stability in the fossil record. How can the diversification that occurred after 40 million years BEST be described? Ⓐ Incomplete, because few fossils remain to show the result of rapid changes in the species Ⓑ Stasis, because the species remained unchanged for long periods of time owing to a stable environment Ⓒ Gradualism, because well-preserved transitional fossils detail that the species evolved steadily over time Ⓓ Punctuated equilibrium, because the species was stable and then underwent rapid changes resulting in new species *Correct Answer (D) ^highly chosen incorrect answer (B)	Data Analysis	
	Item	State Local
	A	
	B	^
	C	
	D*	51
	Item Type	Multiple Choice
	Points	1
	Descriptive Measures	
	- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns	

Evolutionary Theory: Evolution

B.9(B) examine scientific explanations for varying rates of change such as gradualism, abrupt appearance, and stasis in the fossil record (Readiness)

2026 – Q21

Researchers have uncovered a fossil of a *Ginkgo biloba* plant that is estimated to be 165 million years old. This specimen has observable traits, such as fan-shaped leaves, that make it appear like *G. biloba* plants that are found in China today.

Which statement **BEST** explains the stasis of *G. biloba* plants over millions of years?

(A) Periods of glaciation followed by warm climates resulted in disturbances to the *G. biloba* species.

(B) The *G. biloba* lineage is highly resilient to environmental changes, resulting in few disturbances to the species.

(C) A series of volcanic eruptions with alternating coverage of ash and lava resulted in few disturbances to the *G. biloba* species.

(D) *G. biloba* plants evolved in an area that is covered by an inland sea, resulting in disturbances to the species.

*Correct Answer (B)

^highly chosen incorrect answer (C)

Data Analysis

Item	State	Local
A		
B*	64	
C	^	
D		
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

Item Complexity

- Basic application
- Think critically
- Think critically in complex contexts

Stimulus

Academic Vocabulary

Common Error Patterns

Evolutionary Mechanisms

B.10 Biological evolution. The student knows evolutionary theory is a scientific explanation for the unity and diversity of life that has multiple mechanisms.

Evolutionary Mechanisms: Natural Selection

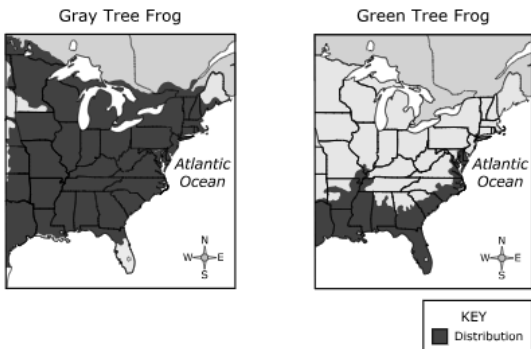
B.10(A) analyze and evaluate how natural selection produces change in populations and not in individuals (Supporting)

2026 – Q27

The table compares the characteristics of gray and green tree frogs.

	Gray Tree Frogs	Green Tree Frogs
Habitat	Wooded areas	Wetlands
Food	Spiders, snails, small insects	Flies, mosquitoes, other small insects
Predators	Snakes, birds, large fish	Snakes, birds, large fish
Appearance		

The maps indicate where gray and green tree frogs are found.



Which statement **BEST** explains how natural selection has allowed both species to evolve and survive in overlapping areas?

- (A) Each individual adapted to have traits that allowed the two species to live in different locations.
- (B) Each population adapted to have similar roles within the same ecosystem.
- (C) Each individual adapted to have different coloration, allowing each species to blend into a specific environment.
- (D) Each population adapted to eat different species, allowing the two populations to exist without competing.

***Correct Answer (D)**
^highly chosen incorrect answer (C)

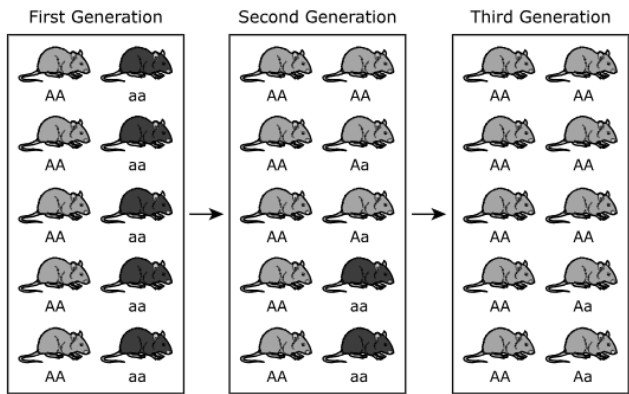
Data Analysis		
Item	State	Local
A		
B		
C	^	
D*	21	
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Evolutionary Mechanisms: Natural Selection

B.10(B) analyze and evaluate how the elements of natural selection, including inherited variation, the potential of a population to produce more offspring than can survive, and a finite supply of environmental resources, result in differential reproductive success (Readiness)

2026 – Q10

This question has two parts.
The diagram shows changes in the gene pool of a mouse population over time.



Part A
Which conclusion is best supported by the diagram?

- ☐ A Mice with the gray phenotype can survive only when they are homozygous dominant.
- ☐ B Mice with the gray phenotype have higher reproductive success than mice with the black phenotype do.
- ☐ C Mice with the black phenotype changed to the gray phenotype before reproducing.
- ☐ D Mice with the black phenotype are more fit for their environment than mice with the gray phenotype are.

Part B
Which evidence supports the answer to Part A?

- ☐ A The percentage of mice that are gray is increasing in each generation.
- ☐ B Some mice in the third generation still have the recessive allele.
- ☐ C The same 10 individuals are shown in all three generations.
- ☐ D The allele frequency was the same for all three generations.

*Correct Answer (B, A)

Data Analysis		
Item	State	Local
Full Credit	68	
No Credit	23	
Partial Credit	8	
Item Type	Multipart	
Points	2	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Evolutionary Mechanisms: Natural Selection

B.10(B) analyze and evaluate how the elements of natural selection, including inherited variation, the potential of a population to produce more offspring than can survive, and a finite supply of environmental resources, result in differential reproductive success (Readiness)

2026 – Q23

The table contains reproductive information about eastern cottontail rabbits.

Reproductive Characteristic	Description
Breeding season	March–September
Litters per breeding season	3–8
Litter size	3–6
Maximum offspring per breeding season	48

Which statement **BEST** explains how a reproductive characteristic of eastern cottontail rabbits allows the species to undergo natural selection?

- ☐ (A) Having many offspring ensures that some survive predation.
- ☐ (B) Multiple litters ensure that parents do not have to care for offspring.
- ☐ (C) Large litter size ensures that offspring are adapted to the environment.
- ☐ (D) Short breeding seasons ensure that resources are available to offspring.

*Correct Answer (A)
^highly chosen incorrect answer (C)

Data Analysis

Item	State	Local
A*	55	
B		
C	^	
D		
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

- Item Complexity
 - Basic application
 - Think critically
 - Think critically in complex contexts
- Stimulus
- Academic Vocabulary
- Common Error Patterns

IQ Analysis Investigating the Question	B.10(C)	RC 3
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Evolutionary Mechanisms: Natural Selection

B.10(C) analyze and evaluate how natural selection may lead to speciation (Readiness)

2026 – Q17

Peppered moths are widely found across Britain. Traditionally, the most common phenotype of the moth has white wings with black speckles. This pattern provides camouflage for moths resting on grayish tree trunks. Another phenotype of the moth has dark, almost black, wings. These darker moths are not as well camouflaged on gray tree trunks. During the Industrial Revolution, air pollution increased in and around cities, and soot blackened the trunks of trees.

Which statement **MOST LIKELY** identifies how the Industrial Revolution affected the population of peppered moths in British cities?

- Ⓐ Moths with darker wings likely became more common, because natural selection favors the passing on of traits that help an organism survive and can lead to the formation of a new species.
- Ⓑ Moths with white wings likely became more common because moths with mutations die before reproducing.
- Ⓒ Moths with darker wings likely became less common because natural selection by predators can lead to extinction.
- Ⓓ Both moths with white wings and moths with darker wings likely became more common, because natural selection leads to the divergence of species into types that do not compete with each other for resources.

*Correct Answer (A)
^highly chosen incorrect answer (C)

Data Analysis		
Item	State	Local
A*	71	
B		
C	^	
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Evolutionary Mechanisms: Natural Selection

B.10(C) analyze and evaluate how natural selection may lead to speciation (Readiness)

2026 – Q29

Hummingbirds have long beaks that allow them to drink nectar from flowers. Hummingbirds were originally present only at the base of the Andes mountains, but their territory has expanded over time. Some hummingbirds developed longer beaks that enabled them to drink nectar from larger flowers that grow at higher altitudes.

Which statement **BEST** explains how natural selection affected hummingbirds?

A The level of genetic diversity decreased because food sources became scarce.

B The number of heritable traits decreased even though food sources were plentiful.

C An adaptation provided some individuals better access to food resources, causing the original population to split into two species.

D The original population lost genes due to migration, increasing competition for food sources.

* Correct Answer (C)

^highly chosen incorrect answer (B)

Data Analysis

Item	State	Local
A		
B	^	
C*	79	
D		
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

Item Complexity

- Basic application
- Think critically
- Think critically in complex contexts

Stimulus

Academic Vocabulary

Common Error Patterns

IQ Analysis Investigating the Question	B.10(D)	RC 3
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Evolutionary Mechanisms: Nonadaptive Mechanisms

B.10(D) analyze evolutionary mechanisms other than natural selection, including genetic drift, gene flow, mutation, and genetic recombination, and their effect on the gene pool of a population (Supporting)

2026 – Q3

A strong storm blows one dark-colored beetle to an area that contains only light-colored beetles of the same species. The environment of the light-colored beetles contains many different kinds of plants, both light and dark in color variation.

What is the **MOST LIKELY** outcome of this drift into the beetle population?

A

All beetles will become light colored over time.

B

All beetles will become dark colored over time.

C

The beetles will become a mix of light and dark colors over time.

D

The beetles will acquire light bodies with dark wings over time.

*Correct Answer (C)

^highly chosen incorrect answer (A)

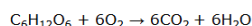
Data Analysis				
Item	State	Local		
A	^			
B				
C*	85			
D				
Item Type	Multiple Choice			
Points	1			
Descriptive Measures				
• Item Complexity - Basic application - Think critically - Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns				

Energy Conversions in Organisms

B.11 Biological structures, functions, and processes. The student knows the significance of matter cycling, energy flow, and enzymes in living organisms.

B.11(A) explain how matter is conserved and energy is transferred during photosynthesis and cellular respiration using models, including the chemical equations for these processes (Supporting)

The equation for cellular respiration is shown.



What happens to oxygen during cellular respiration?

- ☐ (A) Oxygen atoms are used as a source of energy.
- ☐ (B) Oxygen atoms are released into the atmosphere.
- ☐ (C) Oxygen atoms are rearranged in molecules.
- ☐ (D) Oxygen atoms are split.

Which evidence **BEST** supports the answer to Part A?

- (A) Oxygen atoms combine with hydrogen to form water, which is evidence that the oxygen atoms are conserved.
- (B) Oxygen atoms are used to provide the energy needed for cellular respiration, which means that the oxygen atoms are not conserved.
- (C) Oxygen atoms combine with carbon and hydrogen to form glucose, which is evidence that the oxygen atoms are conserved.
- (D) Oxygen atoms are released so that they can be used by other organisms, which means that the oxygen atoms are not conserved.

***Correct Answer (C, A)**

Local

14

71

15

Multipart

2

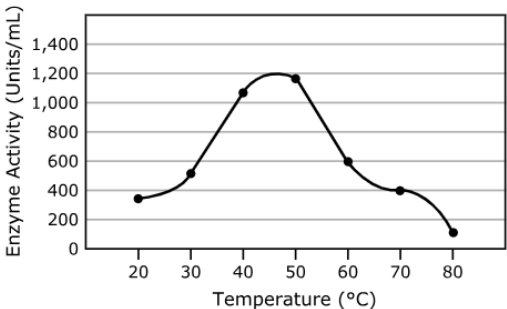
- **Item Complexity**
 - Basic application
 - Think critically
 - Think critically in complex contexts
- **Stimulus**
- **Academic Vocabulary**
- **Common Error Patterns**

Energy Conversions in Organisms: Cellular Respiration

B.11(B) investigate and explain the role of enzymes in facilitating cellular processes (Readiness)

2026 – Q7

A graph of enzyme activity is shown.



Source: Jayashree, S., et al., *Journal of Genetic Engineering and Biotechnology*, 2014

Which statement is **BEST** supported by the data?

- (A) The enzyme is denatured and can no longer bind to its substrate between 20 and 40 °C.
- (B) The enzyme is able to optimally function and bind to its substrate at around 45 °C.
- (C) The enzyme can only bind to its substrate at temperatures between 30 and 60 °C.
- (D) The enzyme can no longer bind to its substrate at temperatures greater than 60 °C.

*Correct Answer (B)
^highly chosen incorrect answer (C)

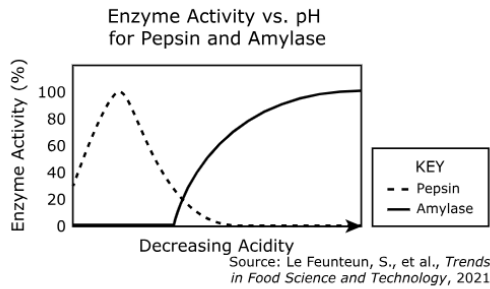
Data Analysis		
Item	State	Local
A		
B*	55	
C	^	
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Energy Conversions in Organisms: Cellular Respiration

B.11(B) investigate and explain the role of enzymes in facilitating cellular processes (Readiness)

2026 – Q24

The graph shows the reaction rates for two different enzymes at different pH levels.



Each enzyme is used in the body for different reactions:

- Pepsin is used in the stomach to help digest proteins.
- Amylase is used in the mouth to help digest sugars.

Which statement **BEST** describes what will happen when amylase is swallowed and enters the stomach?

- ☐ A Amylase will continue to break down sugars in the stomach.
- ☐ B Amylase will help the stomach break down proteins.
- ☐ C Amylase will be recycled for use in other systems.
- ☐ D Amylase will be unable to function in the stomach.

*Correct Answer (D)
^highly chosen incorrect answer (A)

Data Analysis		
Item	State	Local
A	^	
B		
C		
D*	35	
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

Interactions Among Biological Systems

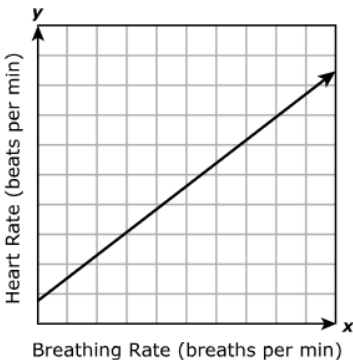
B.12 Biological structures, functions, and processes. The student knows that multicellular organisms are composed of multiple systems that interact to perform complex functions.

Interactions Among Biological Systems: Animal Structure and Function

B.12(A) analyze the interactions that occur among systems that perform the functions of regulation, nutrient absorption, reproduction, and defense from injury or illness in animals (Supporting)

2026 – Q2

Students are studying the relationship between heart rate and breathing rate as a person exercises. They make the graph shown.



Based on the graph, the students can conclude that which pair of body systems interact during exercise?

- ☒ A Circulatory and respiratory systems
- ☐ B Digestive and integumentary systems
- ☐ C Endocrine and nervous systems
- ☐ D Circulatory and immune systems

*Correct Answer (A)

Data Analysis		
Item	State	Local
A*	87	
B		
C		
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

IQ Analysis Investigating the Question	B.12(B)	RC 1
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Interactions Among Biological Systems: Plant Structure and Function

B.12(B) explain how the interactions that occur among systems that perform functions of transport, reproduction, and response in plants are facilitated by their structures (Readiness)

2026 – Q13

Based on Figure 1, explain how plant roots support optimal levels of nutrients for the entire plant.

Choose the correct answer from each drop-down box to complete the sentences.

The roots have high , which enables the of nitrates transported by to other parts of the plant. One effect of fungi is in structural components that contribute to proper root functioning.

*Correct Answer (surface area, absorption, xylem, an increase)

Data Analysis		
Item	State	Local
Full Credit	19	
No Credit	16	
Partial Credit	65	
Item Type	Inline Choice	
Points	2	
Descriptive Measures		
- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

Interactions Among Biological Systems: Plant Structure and Function		
B.12(B) explain how the interactions that occur among systems that perform functions of transport, reproduction, and response in plants are facilitated by their structures (Readiness)		
2026 – Q33 Which statement BEST describes the purpose of gravitropism in plants? Ⓐ Upward growth of the stem reduces surface area, while downward growth of the roots maximizes transfer of nutrients. Ⓑ Upward growth of the stem maximizes photosynthesis, while downward growth of the roots maximizes water absorption. Ⓒ Upward growth of the stem maximizes water absorption, while downward growth of the roots maximizes the spread of seeds. Ⓓ Upward growth of the stem reduces predation, while downward growth of the roots maximizes the production of hormones. *Correct Answer (B) ^highly chosen incorrect answer (A)	Data Analysis	
	Item	State Local
	A	^
	B*	65
	C	
	D	
	Item Type	Multiple Choice
	Points	1
	Descriptive Measures	
	- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns	

Interdependence and Ecosystem Stability

B.13 Interdependence within environmental systems. The student knows that interactions at various levels of organization occur within an ecosystem to maintain stability.

Interdependence and Ecosystem Stability: Ecosystem Stability

B.13(A) investigate and evaluate how ecological relationships, including predation, parasitism, commensalism, mutualism, and competition, influence ecosystem stability (Supporting)

2026 – Q31

Endophytes are microorganisms that live between the cells of plant tissues. The data table shows characteristics of rice plants in the presence and absence of different species of endophytes.

Endophyte	Plant Shoot Height (cm)	Grain Yield (tons of rice/hectare)
Bacteria Species W	47.5	2.89
Mixture of Bacteria Species W, X, Y, Z	60.2	3.07
None	42.1	2.80

Source: Govindarajan, M., et al., *Microbial Ecology*, 2008

What claims can be made about endophytes based on the data in the table?

Move the correct answer to each box. Not all answers can be moved to all boxes.

enhances

impairs

symbiotic relationship

parasitic relationship

The presence of certain microorganisms the health of rice plants. The difference in plant characteristics in the presence and absence of bacterial endophytes supports a claim that the rice plants and microorganisms are part of a

.

*Correct Answer (enhances, symbiotic relationship)

Data Analysis

Item	State	Local
Full Credit	65	
No Credit	35	
Item Type	Drag and Drop	
Points	1	

Descriptive Measures

Item Complexity

Basic application

Think critically

Think critically in complex contexts

Stimulus

Academic Vocabulary

Common Error Patterns

https://lead4ward.com/resources/#tab-374c424d471c2eba993

40/45

B.13(B) analyze how ecosystem stability is affected by disruptions to the cycling of matter and flow of energy through trophic levels using models (Supporting)

The sea otter is a keystone species in the eastern Pacific Ocean, responsible for maintaining ecosystem health. In California's Monterey Bay, healthy sea otter populations are linked to lush, productive kelp forests. Sea urchins are the primary prey of sea otters, and the urchins feed on the kelp.

Kelp → Sea urchins → ~~Sea otters~~

Think about the information carefully. Then enter your response in the box provided.

B *I* U *T_x* := :: ≡ ≡ | ✂ 📄 🗑️ ↶ ↷ Ω

Chars 0/47

***Correct Answer (See Scoring Guide)**

2

- **Item Complexity**
 - Basic application
 - Think critically
 - Think critically in complex contexts
- **Stimulus**
- **Academic Vocabulary**
- **Common Error Patterns**

IQ Analysis Investigating the Question	B.13(C)	RC 4
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Interdependence and Ecosystem Stability: Ecosystem Stability				
B.13(C) explain the significance of the carbon and nitrogen cycles to ecosystem stability and analyze the consequences of disrupting these cycles (Supporting)				
2026 – Q12 Based on Figure 1 and Table 1, explain how a fire affects soil and the nitrogen cycle. Move the correct answer to each box. Each answer may be used more than once. Not all answers will be used. Not all answers can be moved to all boxes. ammonium nitrate nitrite an increase a decrease Because numbers of nitrifying bacteria decrease after a fire, there is a buildup of ions that cannot be converted into ions and then into ions. The results of this process would likely be in levels of nitrogen in the form that plants can use and in ecosystem stability.		Data Analysis		
		Item	State	Local
		Full Credit	20	
		No Credit	47	
		Partial Credit	34	
		Item Type	Drag and Drop	
		Points	2	
		Descriptive Measures		
		- Item Complexity - Basic application - Think critically - Think critically in complex contexts - Stimulus - Academic Vocabulary - Common Error Patterns		

*Correct Answer (ammonium, nitrite, nitrate, a decrease, a decrease)

Interdependence and Ecosystem Stability: Ecosystem Stability

B.13(D) explain how environmental change, including change due to human activity, affects biodiversity and analyze how changes in biodiversity impact ecosystem stability (Readiness)

2026 – Q16

Chinese tallow trees are ornamental plants that have spread from residential gardens to other areas of Texas. These trees have a deep taproot that allows them to withstand drought and outcompete native plant species for water resources. Their leaves and fruit are toxic to humans and cattle.

Which statement **BEST** describes how Chinese tallow trees affect Texas ecosystem stability?

- ☐ Ⓐ Chinese tallow trees increase ecosystem stability by providing a food source that is hardy and drought-resistant.
- ☐ Ⓑ Increased diversity from the introduction of Chinese tallow trees to Texas provides more stability to the ecosystem.
- ☐ Ⓒ The presence of Chinese tallow trees decreases biodiversity and ecosystem stability by competing with native plants for resources.
- ☐ Ⓓ The introduction of Chinese tallow trees did not affect the stability of the ecosystem because the total number of plants present to perform photosynthesis remains the same.

***Correct Answer (C)**
^highly chosen incorrect answer (B)

Data Analysis		
Item	State	Local
A		
B	^	
C*	64	
D		
Item Type	Multiple Choice	
Points	1	
Descriptive Measures		
• Item Complexity ◦ Basic application ◦ Think critically ◦ Think critically in complex contexts • Stimulus • Academic Vocabulary • Common Error Patterns		

Interdependence and Ecosystem Stability: Ecosystem Stability

B.13(D) explain how environmental change, including change due to human activity, affects biodiversity and analyze how changes in biodiversity impact ecosystem stability (Readiness)

2026 – Q30

In tropical ecosystems, some agricultural crops can be grown under existing trees. Other crops require trees to be removed in order for the crops to grow.

Studies have found that removing trees prior to planting crops affects the biodiversity of the ecosystem differently from planting crops under trees.

Which statement **BEST** explains the impact that removing trees prior to planting crops has on the biodiversity of the tropical ecosystem?

- (A) Removing trees reduces biodiversity because crops do not provide energy for predators.
- (B) Removing trees reduces biodiversity because trees provide both food and habitat for animals.
- (C) Removing trees has no impact because crops need only one type of insect in order to be pollinated.
- (D) Removing trees increases biodiversity because the trees are replaced by different types of crops.

*Correct Answer (B)

^highly chosen incorrect answer (D)

Data Analysis

Item	State	Local
A		
B*	70	
C		
D	^	
Item Type	Multiple Choice	
Points	1	

Descriptive Measures

- **Item Complexity**
 - Basic application
 - Think critically
 - Think critically in complex contexts
- **Stimulus**
- **Academic Vocabulary**
- **Common Error Patterns**

[Cluster/Genre]: [Subcluster]

[Student Expectation] [type]

Data Analysis

Item	State	Local
Item type		
Points		

Descriptive Measures

- **Item Complexity**
 - Basic application
 - Think critically
 - Think critically in complex contexts
- **Stimulus**
- **Academic Vocabulary**
- **Common Error Patterns**

*Correct Answer

